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**SOCIAL PRESENCE AND EMOTIONAL INTERACTION:
COMPARING NAO ROBOT AND ITS DIGITAL TWIN**

Introduction and State of the Art

The field of Human-Robot Interaction (HRI) is currently shaped by an ongoing and open debate: whether a virtual robot can serve as a valid substitute for its physical counterpart in experimental studies. The use of physical robots, like the widely used NAO humanoid, entails significant costs and logistical complexities that can limit the scope and accessibility of research.

The existing literature presents a conflicted view. Some studies, such as Li et al. (2019) [1], have shown that virtual robots can be perceived as more "discomforting," with humans maintaining a larger personal distance compared to physical agents. This suggests that physical presence is crucial for establishing social norms and comfort. Conversely, other work, including studies by Thellman et al. (2016) [2], indicates that for certain tasks, a robot's physical form may not be the primary driver of the interaction's success. This perspective is strongly supported by a recent meta-analysis from Esterwood et al. (2025) [3], which synthesized findings from numerous studies and found no significant overall differences in key social metrics like perceived anthropomorphism, social presence, or user engagement. The tasks reviewed in their work often included collaborative problem-solving, educational tutoring, and persuasive interactions, where the agent's functional behavior was more critical than its physical instantiation. These findings suggest that the quality of the interaction and the agent's perceived social capability may, in many contexts, be more influential than its physical form.

While factors like proxemics, trust, and social influence have been compared, there has been a lack of focused, quantitative investigation into how embodiment specifically affects the human perception of non-verbal emotional cues, that is the aim of this thesis. This is particularly relevant for robots like the NAO, which convey emotions primarily through body language rather than complex facial expressions. The central research question is therefore: Does the physical presence of a social robot fundamentally alter how humans recognize its emotional expressions compared to a visually and kinematically identical digital twin?

To ground this investigation, the thesis draws upon two complementary emotional models: Paul Ekman's theory of the six basic, discrete emotions [4] to define what is being expressed, and James Russell's circumplex model of affect [5] to understand how these expressions are perceived and interpreted dimensionally.

Motivation

To address this gap, this thesis pursued two interconnected objectives:

- **Primary Objective:** To empirically investigate whether the physical presence of the NAO robot influences the accuracy with which humans recognize its emotional expressions, comparing it to the accuracy achieved with a visually and kinematically identical digital twin. The goal was to contribute to the embodiment debate with quantitative data, determining whether virtual simulations can be considered a valid alternative for research in the affective domain.
- **Secondary Objective:** To design and implement a novel software framework using an empathetic Large Language Model (LLM) to manage real-time dialogue enriched with emotional expressions. This framework had to be operational on both the physical and virtual platforms, serving as a proof-of-concept for future, more complex HRI interactions and as a tool for rapid prototyping of social behaviors.

Methods and Procedure

A rigorous methodology was implemented to achieve the objectives. First, a high-fidelity digital twin of the NAO robot was created. Starting from the original CAD files, a 3D model was developed in Blender, equipped with a skeleton (rig) with the same 25 degrees of freedom as the physical robot. This model was then imported into the Unity game engine and made fully articulable in real time.

Subsequently, a unified software architecture was developed to ensure behavioural parity between the two robots. A central server, based on Flask, orchestrated the experiment by sending the exact same motion commands to two distinct control pathways: one for the physical robot (via the pynaoqi library) and one for the virtual robot (via a TCP socket to Unity). This design isolated physical embodiment as the sole independent variable.

The experimental design was built upon the two chosen emotional models. First, a set of 12 emotional gestures was created, representing three distinct variations for each of four basic, discrete emotions selected from Paul Ekman’s universally recognized set: Happiness, Sadness, Anger, and Fear. These gestures were systematically adapted from validated studies in the HRI literature.

A within-subjects, counterbalanced experiment was conducted with 24 participants. After each gesture, participants performed two tasks. First, they performed a classification task, identifying the gesture from the four Ekman emotions. Second, to capture a more nuanced understanding of their perception, they provided a dimensional rating of the emotion’s Valence (pleasantness) and Arousal (intensity) using the 9-point Self-Assessment Manikin (SAM) scales, a tool directly derived from Russell’s circumplex model of affect. This dual-measurement approach allowed for both an analysis of recognition accuracy and an in-depth exploration of the underlying perceptual space.

Results and Discussion

The results of the study delivered a compelling answer to the research question. The primary finding was a strong null effect: there was no statistically significant difference in emotion recognition accuracy between the physical robot condition (77.8% accuracy) and the virtual robot condition (75.7% accuracy). A repeated-measures ANOVA confirmed that embodiment was not a significant factor in how well participants could identify the robot’s emotion.

Instead, the analysis revealed that the most dominant factor was the intrinsic clarity of the gesture itself. The type of emotion being expressed had a significant effect on recognition success. Happiness, with its culturally unambiguous and distinct gestures, was recognized with over 90% accuracy. In contrast, the negative-valence emotions—sadness, anger, and fear—were frequently confused with one another. The dimensional analysis using SAM data provided a clear explanation: participants correctly identified the negative valence of these emotions but struggled to differentiate them based on their perceived arousal levels. This finding shifts the focus of the embodiment debate: for successful emotional communication in this context, the quality of gesture design is more critical than the robot’s physical or virtual form.

This quantitative finding was enriched by qualitative feedback. Five participants noted that interacting with the physical robot felt "more comfortable", further studies should explore this perception, for example using the RoSas scale [6]. This highlights a crucial distinction between functional and experiential equivalence. While the virtual

twin is functionally a valid substitute for the task of emotion recognition, the physical robot engenders a more positive subjective experience.

This distinction was further explored in an exploratory study of the LLM-driven empathetic dialogue framework. The study involved 4 persons, 2 interacted with NAO, 2 with the digital twin, at the end of the interaction the user filled a RoSas (Carpinella, 2017) questionnaire [6]. Despite being driven by identical LLM logic, the physical robot was perceived as more competent, while the low Discomfort is perceived as the same between the robots, suggesting that physical presence confers a psychological "aura of capability". This demonstrates that while a digital twin is an invaluable tool for prototyping and testing specific functionalities like gesture recognition, the subtle but powerful experiential effects of physical embodiment cannot be ignored in applications targeting long-term social engagement.

Conclusion and Future Work

The aim of this thesis was to investigate the role of physical embodiment in Human-Robot Interaction by comparing human perception of emotional expressions from a physical agent versus its digital counterpart. To provide a definitive answer, a comprehensive experimental platform was constructed. The cornerstone of this work was the development of a high-fidelity, kinematically identical digital twin of the NAO robot, meticulously modeled and rigged to ensure perfect visual and behavioral parity with its physical counterpart. A unified software architecture was then engineered as the core of the study, capable of dispatching the exact same motion commands to both the physical robot and its virtual agent. This setup allowed for the deployment of 12 distinct emotional gestures, systematically ensuring that physical embodiment was the sole independent variable under investigation.

The primary conclusion is that physical embodiment has no significant effect on the accuracy of emotional gesture recognition. Instead, the legibility of the gesture is the determining factor. This validates the use of high-fidelity digital twins as a scalable, cost-effective, and reliable tool for HRI research, particularly for prototyping and testing non-verbal social cues.

Furthermore, a framework for more complex, multimodal interactions has been integrated using a Large Language Model (LLM) to drive empathetic dialogue. An exploratory study using this framework suggests that while empathy is perceived similarly between the two agents, the physical robot was perceived as more competent. This highlights a powerful psychological effect of embodiment that transcends mere functional parity. Future work will build directly on these preliminary findings to deepen the investigation into the perceptual differences between physical and digital agents in rich, interactive contexts. By re-evaluating embodiment in a constrained, multimodal task—such as a game of chess where the robot combines strategic dialogue with emotional expressions—we can explore if the experiential comfort and perceived competence of physical presence translate into measurable behavioral outcomes, thus further clarifying the nuanced relationship between a robot’s form and its function.

References

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